

Title: Remove Bad Channels

Category: Preprocessing

Libraries: Numpy, Pandas, MNE

1. Objective: To identify and repair faulty EEG sensors such as those that are loose (recording static noise) or disconnected (recording nothing). Instead of deleting these sensors, which would create 'holes' in our data, we mathematically repair them using information from neighboring sensors. This ensures the AI model always receives a complete, clean picture of the brain activity.

2. How it Works:

The algorithm operates in two stages: Detection and Repair.

- **Stage A: Detection (Finding the Bad Sensors)**

The algorithm analyzes the entire recording session (not just one row) using two statistical tests:

1. Variance Test (Z-Score): Measures the "**stability**" of a channel.

- Formula: $Z = \frac{\text{Variance} - \text{Mean Variance}}{\text{Std Dev of Variance}}$
- If $|Z| > 3.0$ (the parameter), the channel is flagged as bad (too noisy or dead).

2. Correlation Test: Measures "teamwork. (Based on Numpy)

- If a channel has low correlation (< 0.4 (the second parameter)) with its neighbors, it is flagged as recording random noise instead of brain waves.

- **Stage B: Repair (Global Mean Interpolation)**

Once bad channels are identified, we repair them using Global Mean Interpolation. We replace the noisy data with the average signal of all remaining healthy channels at that specific time point (Fixed based on its row only).

- Formula: $x_{new} = \frac{\sum x_{good}}{N_{good}}$

3. Numerical Example:

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
subject id	21	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
channel 1	0.264321	0.357754	0.036332	0.569334	0.457409	0.766514	0.77816	-1.26863	0.533073	4.342115	-6.54819	0.376736	-4.54787	0.150005	-0.90747	0.461362							

Figure 1 Raw Data

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
subject id	21	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
channel 1	0.264321	0.357754	0.036332	0.569334	0.457409	0.766514	0.77816	-1.26863	0.533073	4.342115	-6.54819	0.376736	-4.54787	0.150005	-0.90747	0.461362							

Figure 2 Data after fixing

Step 1: Status Check

The algorithm identified 3 Bad Channels (Ch 12, 13, 15) based on their overall session statistics. It retained 13 Good Channels (Ch 1-11, 14, 16).

*Note: Channels 8 and 11 have negative values (e.g., -6.64). This is considered **Healthy Behavior**, as EEG signals are waves that oscillate above and below zero. They were kept because their variance over time was normal.*

Step 2: The Calculation

To repair the bad channels, we sum the values of the 13 healthy channels at this exact time:

$$Sum = 9.98 (Ch1) + 9.06 (Ch2) + \dots + 9.16 (Ch14) + 5.47 (Ch16)$$

$$Sum \approx 77.3278$$

We then divide by the count of good channels ($N = 13$):

$$\text{Global Mean} = \frac{77.3278}{13} \approx 5.948291$$

Then it changed the values of those 3 channels to the new value (e.g., **5.948291**)

4. Visual Example:

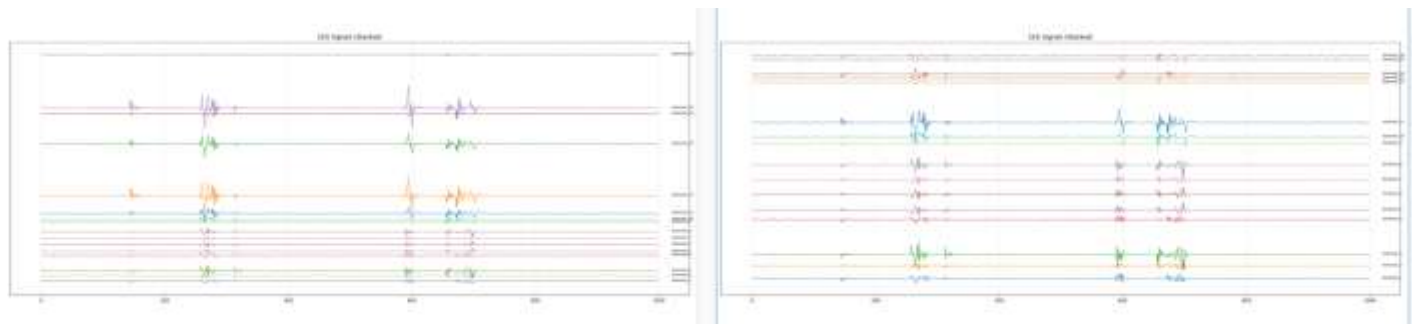


Figure 3 data before vs after